Singha Junchan

Cybersecurity · Cloud-Native Security · Zero-Trust Authorization · AI-Agent Security
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Summary

Final-year Computer Engineering student (**GPA 3.67**) who builds cloud-native and AI security around one idea: *a check cheap enough to run every time, and provable afterwards*, then **proves it on real Kubernetes and AWS**. **Best Paper Award** (JCSSE 2026, second author); first author of papers submitted to IEEE TDSC and accepted at KST 2027. **TU Alert team lead**, **Allianz Ayudhya security intern**, **TOI18 national champion**. Seeking a **cybersecurity internship from January 2027**.

Education

B.Eng. in Computer Engineering – minor in Cloud Computing & Cybersecurity — Aug 2023 – Expected May 2027
Sirindhorn International Institute of Technology (SIIT), Thammasat University — **GPA 3.67 / 4.00** · full scholarship

Research & Engineering

FEDAC – *authorization that adds almost no delay* — Submitted · IEEE TDSC 2026 · first author

- **Designed and built a zero-trust authorization framework**: a hub **verifies each caller once** and issues a short-lived capability, and a **sidecar beside every workload enforces it locally**, with **no network call on the request path**.
- **Four risk-adaptive authentication tiers** (OIDC/JWT, TOTP, risk score, Groth16 ZK proof) and **blockchain-anchored audit**, in Go and Rust.
- **Benchmarked on AWS EKS** against three re-implemented frameworks: **1.4 ms median latency**, about **7,100 requests/s**. Code: github.com/singhaju/FEDAC.

U-HAP – *Kubernetes authorization, compiled* — **Best Paper Award** · IEEE JCSSE 2026 · second author

- **Co-developed a Kubernetes authorization webhook** that lets teams **write ACL, RBAC, ABAC and deny policies in one language**; a compiler turns them into indexes, so decisions take **near-constant time**.
- **Validated against OPA**: updates **1.6–2.7× faster**, **100/100 identical decisions**. Accepted; I presented it at JCSSE 2026. Code: github.com/singhaju/U-HAP.

ZKAudit and ZK-ALBA – *private, provable audit of cloud logs* — Two papers · first author

- **ZKAudit** (**accepted, KST 2027**) and **ZK-ALBA** (journal, in preparation): an auditor **proves** credentials, scope and policy freshness **without revealing them**; replay-safe verifier shards, Merkle commitments (Rust, Amazon EC2).

Four further manuscripts — 2 submitted · 2 in preparation · first author

- **Submitted: PQ-SCVA-MCK** (post-quantum multi-cluster Kubernetes authorization) and **SWARM-Fog** (self-healing industrial IoT). **In preparation: FRESCO** (fresh state for scaling cloud apps) and a SIEM that works on encrypted logs.

MoCA – *AI agents that stay safe even when prompt injection succeeds* — In preparation · journal · first author

- A hijacked agent cannot exceed its **signed human delegation**: checked at an **MCP gateway** and at the **resource**.

Experience

Team Lead & Developer, TU Alert — 2026 – Present
Thammasat University (client project) — *Launching soon*

- **Lead a two-person student team** (we both code) building a LINE (LIFF) incident-reporting system that **routes each report to its office**: TypeScript, Hono, Bun, Prisma, Redis, Next.js; **Semgrep** and **gitleaks** in CI.

Information Security Intern (summer training, CSS300) — Jun – Jul 2026
Allianz Ayudhya — *Bangkok, Thailand*

- **Built two AI-agent prototypes for third-party risk management**: a **supplier-side agent** that answers due-diligence questionnaires with **evidence and confidence scores**, and a **buyer-side agent** that guides business owners.
- **Demonstrated both to the Information Security team** (6 Jul) and closed with a final presentation (31 Jul 2026).

Founder, Curriculum Author & Lead Instructor — 2022 – Present
TOI Tutor and ABIG Center (online) — *toitutor.in.th*

- **Founded and lead two online platforms** that coach for the national olympiads (TOI, POSN); **authored the C++ curriculum** (COM100, COM300) for hundreds of students.

R&D Intern, Amity Solutions – *chatbots, data analysis, prompt engineering, NLP and LLMs* — Jul – Aug 2023

Security Practice & CTF

- **NCSA CTF Thailand Tournament 2026** (Aug 2026): **competed in a team** on web, crypto, forensics, OSINT and mobile challenges.
- **Nmap home lab** (Kali) with NSE · **SOSECURE Road to Cybersecurity 2025** · **TryHackMe** and **Hack The Box**.

Skills

Offensive Security:	Nmap, Wireshark, Burp Suite, Metasploit, OSINT, OWASP Top 10
Defensive & Cloud:	Zero Trust, OPA/Rego, Splunk and Wazuh (SOC labs); Kubernetes (RBAC, ABAC, authorization webhooks), AWS (EKS, EC2), Docker, Hyperledger Fabric, Azure basics
Cryptography:	zk-SNARKs (Groth16), Merkle commitments, post-quantum ML-DSA and ML-KEM
Programming:	Python, Go, Rust, C++, C, Java, Bash, TypeScript, SQL
Languages:	Thai (native), English (fluent), Chinese (basic)

Awards, Certifications & Leadership

- **Best Paper Award**, IEEE JCSSE 2026 · **National Champion (1st, Gold)**, Thailand Olympiad in Informatics (TOI18), 2022 · **1st Place**, Image Recognition Programming Challenge (Osaka Inst. of Technology and SIIT), 2025.
- **ISC2 CC**, 2025 · **CompTIA Security+** (exam planned Dec 2026) · DataCamp Azure and AWS, 2025.
- **President, Sub Student Committee 32, SIIT** (2024 – present): **leads a committee representing the student body**.